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Executive Summary

The used smartphone market has emerged as a rapidly growing segment of the global mobile technology industry, characterized by the annual exchange of billions of devices. As consumers increasingly seek affordable and sustainable alternatives to new devices, understanding pricing behavior in the secondary market has become crucial for stakeholders such as retailers, manufacturers, financial institutions, and end-users. The ability to accurately predict and categorize used smartphone prices not only enhances transparency but also presents significant opportunities for operational efficiency and profit optimization within the industry.

This project is designed to establish predictive analytics capabilities as supervised machine language for regressing and classifying used smartphone prices with precision and reliability. The analysis draws upon a dataset encompassing fifteen critical attributes across multiple brands, including Samsung, HTC, Huawei, Infinix, Lenovo, LG, Apple, and many more, spanning devices released between 2013 and 2020. Methodologies (regression and classification) are employed to facilitate price estimation and device categorization into price segments. By harnessing machine learning, the project examines the influence of technical specifications, such as screen size, camera resolution, internal memory, RAM, and battery performance, in conjunction with market-related attributes like device brand, release year, and usage duration. These relationships provide valuable insights into the factors driving pricing outcomes in the used smartphone market.

Predictive models have substantial implications beyond academic research. For businesses, precise pricing predictions facilitate optimal inventory allocation, refined pricing strategies, reduced losses from undervaluation, and increased turnover rates. Retailers and refurbishers gain advantages from improved inventory planning and margin protection, while financial institutions involved in device financing or trade-in programs mitigate risks associated with asset depreciation. Consumers benefit from the models' promotion of fairness and trust by providing transparent and data-driven valuations, enabling better decision-making during device purchases and sales.

Additionally, the project offers market information on the relative importance of various device features in determining value depreciation. This knowledge guides obtaining decisions, refurbishment priorities, and competitive positioning for businesses aiming to strengthen their presence in the secondary market. By identifying features—such as camera quality, memory capacity, or battery—that substantially contribute to value, companies can tailor product offerings and marketing strategies more effectively. Ultimately, this analysis contributes to more efficient market operations, enhanced alignment between supply and demand, and improved decision-making across the used smartphone ecosystem.

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1. Introduction

The quick global turnover of mobile devices has catalyzed the development of a secondary market where used smartphones hold a pivotal role. This market is not only driven by consumer demand for cost-effective alternatives but also influenced by broader trends such as sustainability, electronic waste reduction, and the increasing popularity of trade-in and buyback programs. As the lifespan of smartphones diminishes and new models are introduced at an accelerated pace, the ability to evaluate and forecast the value of used devices has become increasingly crucial for sustaining competitiveness within this dynamic ecosystem.

Although the secondary market presents substantial opportunities, it also presents challenges due to the complicated nature of pricing dynamics. Device value is influenced by a various technical attributes, market conditions, and consumer preferences, rendering simplistic pricing strategies ineffective. Predictive analytics serves as a pathway to navigate this complexity, enabling stakeholders to discern concealed patterns, anticipate price fluctuations, and align strategies with actual market demand. By employing advanced modeling techniques, this project attempts to generate insights that can drive both operational efficiency and informed decision-making within the industry.

2. Business and Analytical Goals

2.1 Business Goals

2.1.1 Business Problems

The used smartphone market encounters several critical challenges that this analysis seeks to address.

- The current pricing mechanisms in the used smartphone market frequently exhibit a lack of standardization, resulting in significant disparities for comparable devices. This situation delays the ability of buyers and sellers to establish fair market values, leading to inefficient transactions, and diminished profit margins.
- Retailers and refurbishment companies face challenges in predicting which device categories will maintain their value and sell promptly. This uncertainty poses risks of overstocking low-demand devices or overlooking opportunities with high-value devices, resulting in elevated holding costs and reduced turnover rates.

2.1.2 Business Goals

Goal 1: Maximizing Profitability through Accurate Pricing

Develop predictive models that reliably estimate the resale value of used smartphones with minimal pricing inaccuracies. By ensuring precise alignment with actual market values, the business can optimize transaction efficiency, enhance customer confidence, and boost overall profitability. Comprehensive impact analysis will further identify the attributes that most significantly influence price retention, facilitating more informed procurement and refurbishment strategies.

Goal 2: Optimizing Inventory Management through Device Segmentation

Develop classification models that categorize devices based on their price retention. Accurate and balanced classification will support effective inventory planning, minimize the costs associated with excess stock, and accelerate device turnover. Insights derived from feature analysis will also guide buying and stocking decisions, ensuring that resources are allocated to devices with enhanced resale prospects.

2.2 Analytical Goals

2.2.1 Analytics Objectives

Build regression model/s that predict the resale value of used smartphones based on their features and market factors. Accurate price estimation reduces pricing errors and supports higher profit margins. Regression outcomes will be directly tied to assessing seller profit/loss by ensuring devices are priced competitively without undervaluing high-demand models.

Develop classification models that segment devices into price retention categories. This segmentation supports inventory planning, allowing businesses to prioritize stocking and refurbishment of devices likely to sell at premium values while minimizing costs tied to slower-moving stock.

2.2.2 Analytics Approach

Phase 1: Data Preprocessing and Exploration

The initial phase will guarantee data readiness for both profitability-oriented regression and inventory-focused classification tasks. It involves assessing data quality, addressing missing values in features, and analyzing zero values to differentiate genuine device specifications from errors. Outliers will be identified and managed to prevent distortion of price predictions. Feature engineering will categorize specifications into meaningful groups and compute brand-level average price retention, ensuring that the inputs are relevant to both pricing accuracy and segmentation. Exploratory data analysis will study statistical properties, correlations with indexes, and distribution across categories to reveal patterns associated with profitability and resale potential.

Phase 2: Model Development Strategy

Regression model/s will serve as a baseline to demonstrate interpretability and feature importance. Advanced model/s will be evaluated to capture complicated relationships and enhance prediction accuracy for normalized used price. This directly supports profit maximization by mitigating mispricing risks.

Classification algorithms will function as baseline models for classification. One of the classification models is effective for identifying groups of smartphones with similar price retention characteristics due to its ability to capture similarity-based groupings of devices. It employs a probabilistic approach that is computationally efficient and interpretable, providing prompt insights into category likelihoods.

Phase 3: Model Validation and Evaluation

In regression model/s, evaluation metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and R-squared are employed to assess pricing accuracy and its impact on potential profitability or loss. Conversely, classification models are evaluated based on metrics like accuracy, precision, recall, and F1-score, which strike a balance across categories and mitigate bias toward any group. These metrics ensure reasonable segmentation for inventory management.

3. Data Preprocessing

3.1 Data Understanding

The dataset contains 3,454 records of smartphone data, each containing 15 attributes that span multiple dimensions of device characteristics and market factors. The data encompasses smartphones from eight distinct brands, with release years ranging from 2013 to 2020, thereby providing a comprehensive overview of the evolution of the used smartphone market.

Dataset Composition:

- Records: 3,454 used smartphone entries
- Attributes: 15 comprehensive features
- Time Span: 8 years (2013-2020) of smartphone market data
- Brand Coverage: Samsung, Apple, Xiaomi, Gionee, Alcatel, Acer, Micromax, and so on

Business Relevance: The multidimensionality of the dataset facilitates comprehensive analysis of factors influencing used smartphone pricing. The inclusion of both technical specifications (camera quality, memory, battery) and market factors (brand, age, days used) establishes the foundation for developing robust predictive models capable of addressing practical business challenges in the used device market.

3.2 Attribute Definitions

1. device_brand (Categorical)

- Definition: Manufacturer brand of the smartphone
- Data Type: String/Categorical
- Values: Samsung, Apple, Asus, HTC, Huawei, LG, so on
- Business Role: Brand reputation exerts a substantial influence on price and consumer perception.
- Analysis Relevance: A pivotal factor in price segmentation.

2. os (Categorical)

- Definition: Operating system of the device
- Data Type: String/Categorical
- Values: Android, iOS, Others, Windows
- Business Role: The operating system significantly impacts device usability, update support, and market appeal.
- Analysis Relevance: Essential for compatibility-driven pricing and market segmentation.

3. screen_size (Continuous)

- Definition: Display screen size measured in centimeters
- Data Type: Float
- Range: 5.08 to 30.71 cm
- Business Role: The primary factor that significantly impacts user experience and pricing.
- Analysis Relevance: A robust indicator of device category and price range.

4. 4g (Binary)

- Definition: 4G network connectivity availability
- Data Type: String (yes/no) [1 = Yes, 0 = No]
- Business Role: A crucial connectivity feature that significantly impacts the marketability of devices.
- Analysis Relevance: Connectivity standards influence pricing strategies and market demand.

5. 5g (Binary)

- Definition: 5G network connectivity availability
- Data Type: String (yes/no) [1 = Yes, 0 = No]
- Business Role: A connectivity feature that signifies the device's advanced technological capabilities.
- Analysis Relevance: Feature impacting pricing classification and ensuring long-term value.

6. rear_camera_mp (Continuous)

- Definition: Rear camera resolution in megapixels
- Data Type: Float
- Range: 0.08 to 48 megapixels
- Business Role: Essential specification for people who takes photos/videos.
- Analysis Relevance: A pivotal feature for price differentiation and quality evaluation.

7. front_camera_mp (Continuous)

- Definition: Front camera resolution in megapixels
- Data Type: Float
- Range: 0.0 to 32 megapixels
- Business Role: Essential for photography and video calling capabilities.
- Analysis Relevance: A secondary camera feature that enhances the overall value of the device

8. internal_memory (Continuous)

- Definition: Internal storage capacity in GB
- Data Type: Integer
- Range: 0.01 to 1024 GB
- Business Role: A crucial specification that significantly impacts the usability and data storage capacity of the device.
- Analysis Relevance: A reliable indicator of device tier and price category.

9. ram (Continuous)

- Definition: Random Access Memory capacity in GB
- Data Type: Integer
- Range: 0.020 to 12 GB
- Business Role: Critical performance specification impacting multitasking capabilities.
- Analysis Relevance: A crucial indicator of device performance tier and price segment.

10. battery (Continuous)

- Definition: Battery capacity in mAh (milliampere-hours)
- Data Type: Integer
- Range: 500 to 9,720 mAh
- Business Role: For ensuring the longevity and user experience of a device.
- Analysis Relevance: Factor in determining the utility and long-term value of a device.

11. weight (Continuous)

- Definition: Device weight in grams
- Data Type: Float
- Range: 69 to 855 grams
- Business Role: A physical attribute that influences the portability and user preference of a device.
- Analysis Relevance: Secondary factor influencing user experience and price positioning.

12. release_year (Discrete)

- Definition: Year when the device was originally released
- Data Type: Integer
- Range: 2013 to 2020
- Business Role: A crucial factor in determining the age and technological relevance of devices.
- Analysis Relevance: The primary factor influencing depreciation rates.

13. days_used (Continuous)

- Definition: Number of days the device has been used
- Data Type: Integer
- Range: 91 to 1094 days
- Business Role: A direct indicator of device wear and tear.
- Analysis Relevance: Used for forecasting price declines and evaluating product condition.

14. normalized_used_price (Target Variable - Continuous)

- Definition: Standardized price index for used device pricing
- Data Type: Float
- Range: 1.537 to 6.619
- Business Role: The primary target variable that represents the market value of used devices.
- Analysis Relevance: The primary outcome variable for regression analysis.

15. normalized_new_price (Reference Variable - Continuous)

- Definition: Standardized price index for new device pricing
- Data Type: Float
- Range: 2.901 to 7.848
- Business Role: Establishing a baseline for calculating value depreciation.
- Analysis Relevance: Essential for the development of derived features and classification targets.

3.3 Missing Data Analysis

3.3.1 Missing Value Assessment:

After comprehensive examination of the dataset, the following missing data patterns were identified:

1. rear_camera_mp: 179 missing values

- Impact: Affects 5.18% of dataset, limiting camera quality analysis.
- Proposed Handling: Impute median rear camera values for devices with similar specifications.
- Justification: Camera specifications are correlated with device depreciation value. Median imputation preserves the distribution while preserving analytical validity.

2. Others: 23 missing values (includes: front_camera_mp, internal_memory, ram, battery, weight)

- Impact: Affects 0.67% of dataset, minimal impact on overall analysis
- Proposed Handling: Impute using median values.
- Justification: Missing value exists in the dataset for multiple columns, but historical context is available. Therefore, mean imputation is the most appropriate statistical method for this isolated case.

3.4 Zero Value Data Analysis

3.4.1 Zero Value Assessment:

After the analysis of the dataset, the following zero data were identified:

1. front_camera_mp: 39 zero values

- Impact: Affects 1.13% of dataset, minimal limitation on the analysis.
- Proposed Handling: Context-dependent handling based on corresponding variables, such as ram and battery. If any of these variables is missing or is zero, the entire row can be removed. This is because it is safe to assume that a modern smartphone does not run without a battery and moreover not have a ram so it is an input error.
- Justification: Zero values signify two distinct scenarios: (a) the absence of front cameras in devices, and (b) the absence of data for devices.

3.5 Exploratory Data Analysis (Basic)

3.5.1 Continuous Variables Analysis:

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew
screen_size	1	3454	13.71	3.81	12.83	13.41	3.69	5.08	30.71	25.63	1.02
rear_camera_mp	2	3275	9.46	4.82	8.00	9.45	7.19	0.08	48.00	47.92	0.76
front_camera_mp	3	3452	6.55	6.97	5.00	5.34	4.45	0.00	32.00	32.00	1.80
internal_memory	4	3450	54.57	84.97	32.00	37.83	23.72	0.01	1024.00	1023.99	5.79
ram	5	3450	4.04	1.37	4.00	4.00	0.00	0.02	12.00	11.98	1.07
battery	6	3448	3133.40	1299.68	3000.00	3010.54	1408.47	500.00	9720.00	9220.00	1.33
weight	7	3447	182.75	88.41	160.00	163.82	29.65	69.00	855.00	786.00	3.23
days_used	8	3454	674.87	248.58	690.50	686.23	255.75	91.00	1094.00	1003.00	-0.34
normalized_used_price	9	3454	4.36	0.59	4.41	4.39	0.54	1.54	6.62	5.08	-0.53
normalized_new_price	10	3454	5.23	0.68	5.25	5.24	0.67	2.90	7.85	4.95	-0.08

Figure 1: Detailed descriptive summary for continuous variable

screen_size:

- Mean: 13.71 cm
- Median: 12.83 cm
- Standard Deviation: 3.81 cm
- Distribution: Right-skewed (skew = 1.02), This indicates that most devices are clustered around smaller screen areas, with some larger flagship models being an exception.

weight:

- Mean: 182.75 grams
- Median: 160.00 grams
- Standard Deviation: 88.41 grams
- Distribution: Highly right-skewed (skew = 3.23), proposing a prevalence of lighter devices, with several heavy outliers.

rear_camera_mp:

- Mean: 9.46 MP
- Median: 8.00 MP
- Standard Deviation: 4.82 MP
- Distribution: Right-skewed (skew = 0.76), significantly indicating concentration around mid-range camera specifications, with high resolution outliers.

front_camera_mp:

- Mean: 6.55 MP
- Median: 5.00 MP
- Standard Deviation: 6.97 MP

- Distribution: Highly right-skewed (skew = 1.80), reflecting the evolution of devices from those without front cameras to modern specifications.

internal_memory:

- Mean: 54.57 GB
- Median: 32.00 GB
- Standard Deviation: 84.97 GB
- Distribution: Extremely right-skewed (skew = 5.79), it is noteworthy that most devices come with modest storage capacities, while premium models offer significantly higher storage capacities.

ram:

- Mean: 4.04 GB
- Median: 4.00 GB
- Standard Deviation: 1.37 GB
- Distribution: Right-skewed (skew = 1.07), demonstrating sustained focus around 4GB standard specifications, with gradual progression to more advanced and high-end options.

battery:

- Mean: 3,133.40 mAh
- Median: 3,000.00 mAh
- Standard Deviation: 1,299.68 mAh
- Distribution: Right-skewed (skew = 1.33), indicating standard battery capacities approximately 3,000mAh, with some high-capacity outliers.

days_used:

- Mean: 674.87 days
- Median: 690.50 days
- Standard Deviation: 248.58 days
- Distribution: Slightly left-skewed (skew = -0.34), it is evident that most devices exhibit extended usage periods with a relatively uniform distribution.

normalized_used_price:

- Mean: 4.36
- Median: 4.41
- Standard Deviation: 0.59
- Distribution: Left-skewed (skew = -0.53), it suggests a concentration of options in higher price segments, with a few lower-priced exceptions.

normalized_new_price:

- Mean: 5.23
- Median: 5.25
- Standard Deviation: 0.68
- Distribution: Nearly normal (skew = -0.08), indicating a balanced pricing strategy across device categories



Figure 2: Distribution of variables

3.5.2 Categorical Variables Analysis:

Brand Distribution:

- Samsung: 9.87% (341 devices) - Global smartphone manufacturer dominance
- Huawei: 7.27% (251 devices) - Premium and mid-range market presence
- LG: 5.82% (201 devices) - Established market presence
- Lenovo: 4.95% (171 devices) - Significant market share
- Apple: 1.13% (39 devices) - Limited presence despite quality market positioning
- Others: 14.53% (502 devices) - Fragmented market with numerous smaller brands
- Remaining Brands: 57.56% (1,986 devices) - Including ZTE (4.05%), Xiaomi (3.82%), Oppo (3.73%), Asus (3.53%), Alcatel (3.50%), Vivo (3.39%), Micromax (3.39%), Honor

(3.36%), HTC (3.18%), Motorola (3.07%), Nokia (3.07%), Sony (2.49%), Meizu (1.80%), and others

Connectivity Distribution:

- 4G Availability: 67.6% (widespread adoption)
- 5G Availability: 4.4% (premium feature in some devices)

Operating System Distribution:

- Android: 93.05% (dominant platform)
- iOS: 1.04%
- Windows: 3.97
- Others: 1.94%

3.5.3 Correlation Plot for Multicollinearity

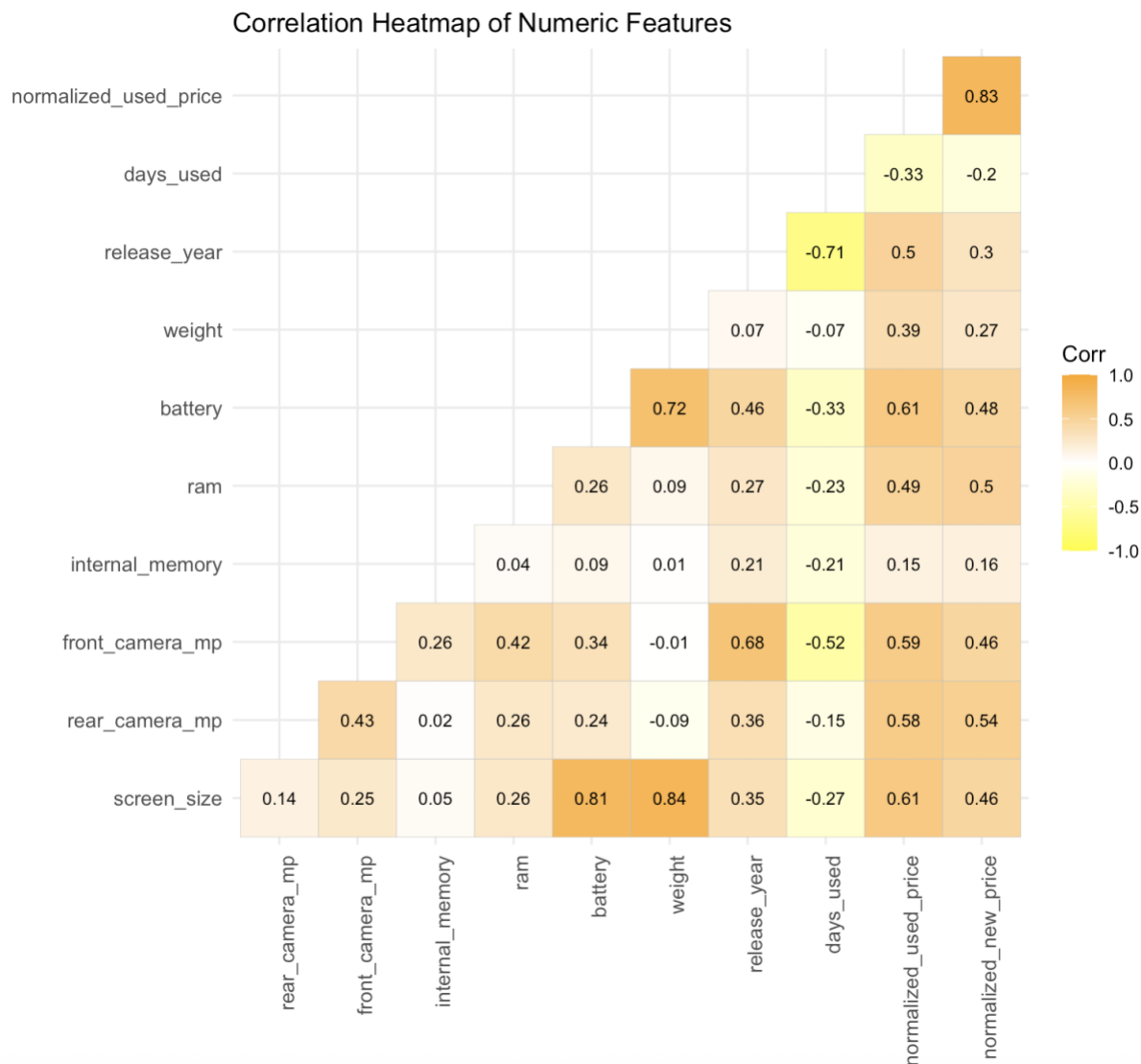


Figure 3: Correlation of Numeric Variables

The correlation heatmap reveals multicollinearity that directly impact both pricing accuracy and inventory management strategies. The strong correlations among attributes; screen size, battery capacity, and weight exhibiting correlations of 0.81, 0.84, and 0.72, respectively; indicate these features function as high end devices rather than independent predictors. This multicollinearity presents both opportunities and risks for achieving the business objectives.

Business Relevancy:

The high correlation (0.83) between normalized used and new prices provides compelling evidence that consistent pricing relationships exist across the smartphone market, supporting the feasibility of standardized pricing algorithms. However, the multicollinearity among other attributes poses significant challenges for model stability and interpretability. When features move together predictably, individual coefficient estimates become unreliable, potentially leading to volatile pricing decisions that could undermine customer confidence and transaction efficiency.

The moderate correlations between camera specifications and pricing (0.58-0.59 for rear cameras, 0.46-0.54 for front cameras) suggest these features serve as reliable independent value drivers, offering a stable foundation for pricing models. Conversely, the weak correlations of internal memory with other features (0.01-0.21 range) indicate storage capacity operates as a distinct value component requiring separate pricing consideration.

The multicollinearity patterns provide invaluable insights for classification models. The clustering of attributes suggests that high end device identification can be simplified by treating screen size, battery, and weight as a composite premium indicator rather than evaluating each individually. This approach streamlines the classification process and diminishes the complexity of inventory decision-making.

3.5.4 Box Plot for Outliers

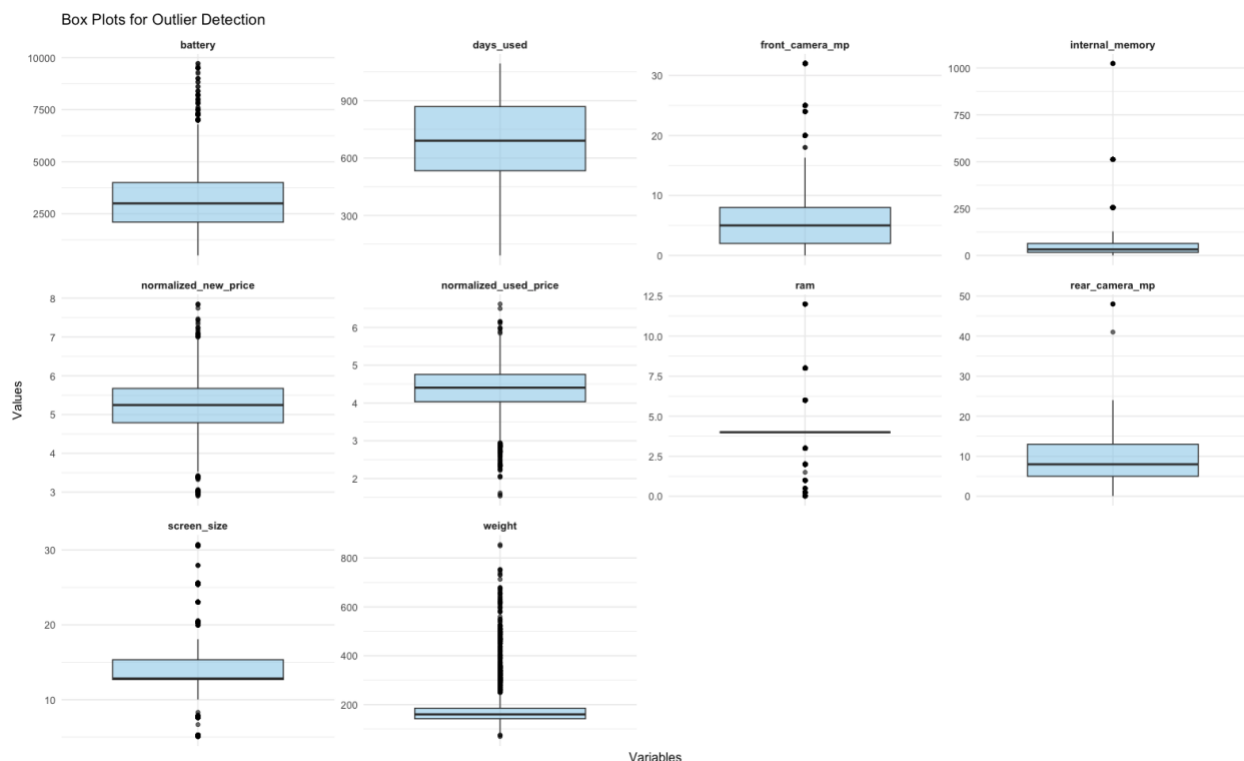


Figure 4: Boxplot for Outlier Detection

This comprehensive boxplot analysis reveals outlier patterns that present both challenges and opportunities for both pricing standardization and inventory optimization within the used smartphone market. The screen size variable exhibits extreme outliers extending beyond 30 centimeters. The camera specifications demonstrate realistic outlier patterns, with rear cameras reaching 48MP and front cameras up to 32MP, indicating flagship devices that command premium prices. These devices could be retained through descriptive statistics rather than being removed as anomalies. Internal memory displays the most concerning outlier pattern, with values up to 1,024GB, while the median remains around 64GB. This suggests either severe market segment bias toward budget devices or data input error that could skew pricing algorithms away from high-value inventory categories. The RAM distribution exhibits extreme concentration at 4GB with minimal outliers, indicating dataset limitations that may not adequately capture modern smartphone specifications and could undermine pricing accuracy for current market conditions. Battery capacity outliers extending beyond 9,000mAh likely means devices that could be handled out by capping the max. Weight outliers above 800g suggest similar data quality issues requiring validation.

Most critically for business applications, the pricing variables exhibit significant outliers in both directions. High outliers represent premium devices that drive profitability, while low outliers indicate either damaged goods or data errors. The days_used variable exhibits a uniform

distribution with minimal outliers, indicating that this factor offers reliable input for depreciation modeling. High-value outliers should be preserved for identifying premium inventory, while extreme outliers should be investigated to address data quality issues that could potentially alter the pricing systems.

4. Predictor Analysis and Relevancy

Predictor analysis ensures the reliability and relevance of variables for constructing predictive models. In the case of this dataset, special attention was given to missing and zero values, as both could change the predictive performance of the models if not addressed properly.

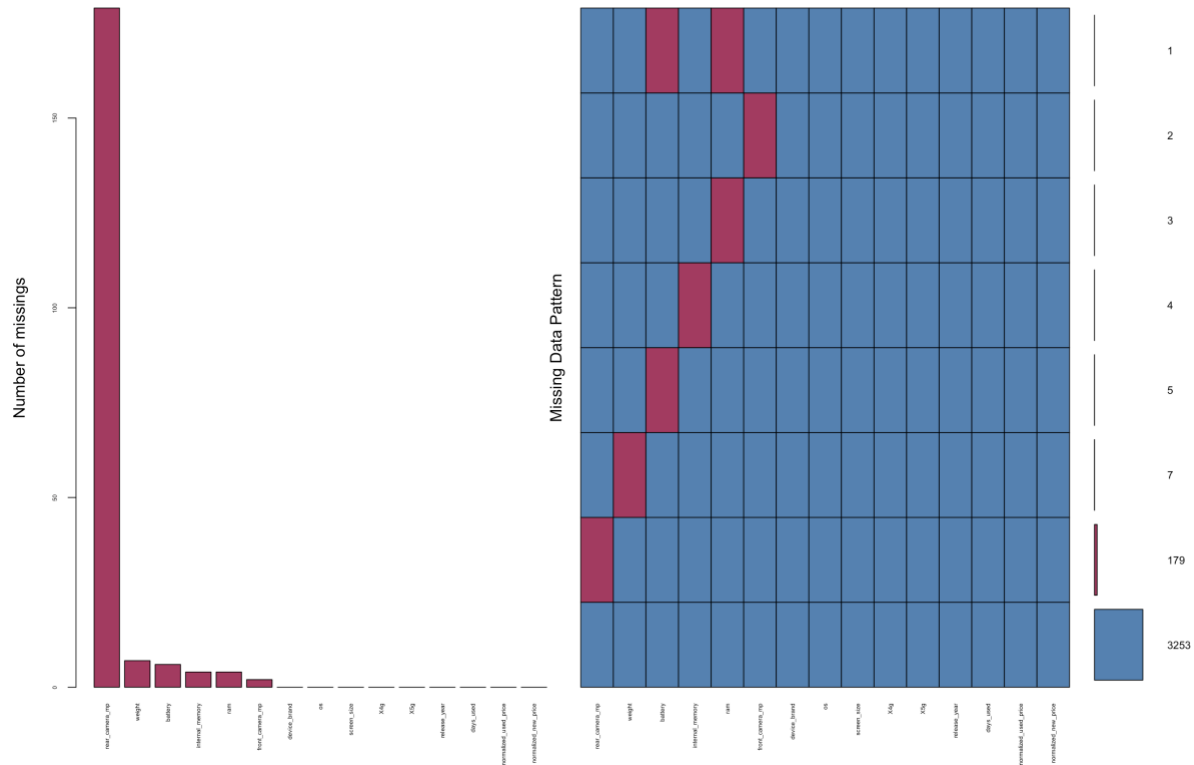


Figure 5: Missing Data Pattern Analysis

4.1 Handling Missing Values

Missing values were encountered across several predictors, including rear and front camera, internal memory, weight, and battery. Given the significance of these features in determining smartphone value, their exclusion was not feasible. Consequently, a structured imputation strategy was used to address these missing values.

- **Brand-Year Median Imputation:** Missing values were imputed by replacing them with the median value of devices from the same brand and release year. This approach is reasonable as manufacturers generally maintain consistent specifications for devices released within a specific year.
- **Brand-Level Median Imputation:** If brand-year information was insufficient (e.g., a limited number of devices were released in that year), the missing value was imputed using the overall brand median. This approach maintains brand-specific tendencies (e.g.,

Apple devices tend to be lighter, while Samsung devices typically have larger batteries on average).

- Global Median Imputation: If neither brand-year nor brand-only medians were available, the global dataset median was utilized. This approach ensured that every record possessed a value, thereby mitigating data loss while minimizing distortion.

This approach aligns with business goals by maintaining accuracy in price prediction, ensuring fair classification across categories, and enabling valid impact analysis on features.

4.2 Handling Zero Values

Some predictors contained values of zero. Although technically valid in certain rare contexts, zero values in this dataset were treated as data entry errors because:

- No modern smartphone has 0 GB internal memory.
- No working smartphone has 0 mAh battery.

To address this issue, missing values were imputed for zero values using the strategy previously mentioned. This approach prevented unrealistic values from distorting statistical analysis while preserving the integrity of the dataset.

Business relevance:

By treating zero values as errors, the analysis prevented the generation of misleading insights, such as suggesting that certain devices are memory deficient. This aligns with the business requirement for accuracy in pricing and classification, ensuring that feature-driven impact analysis accurately reflects real-world devices rather than erroneous data.

4.3 Handling Multicollinearity

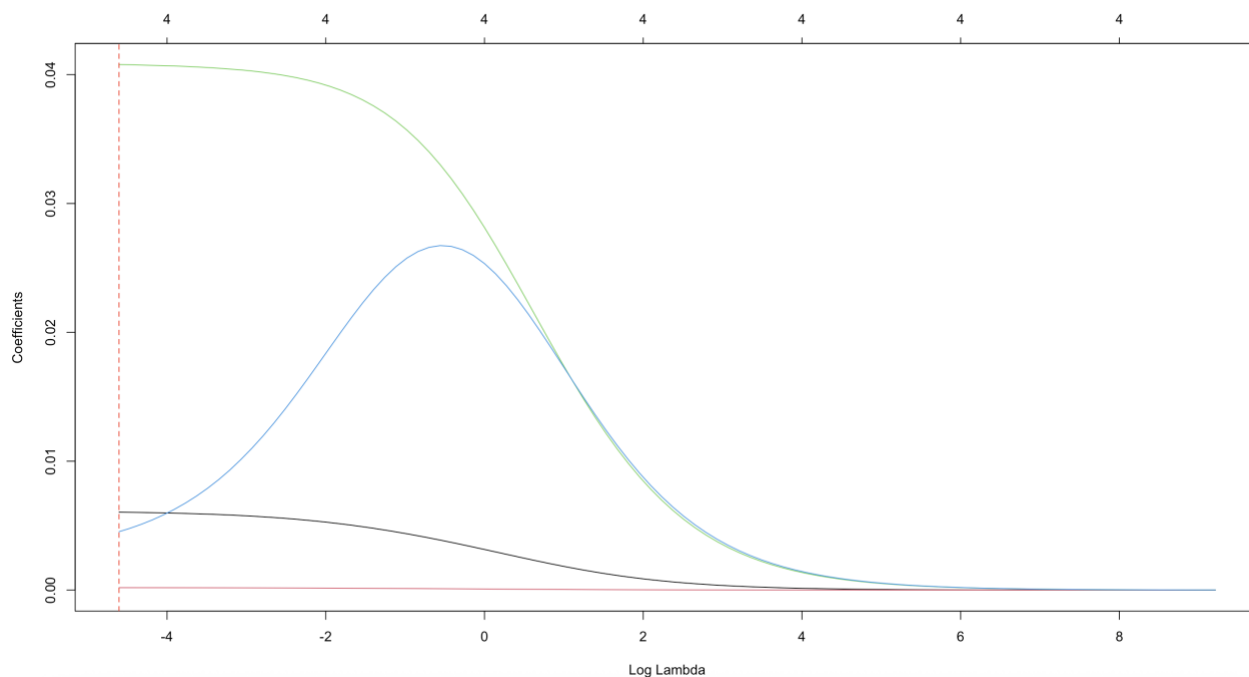


Figure 6: Ridge Shrinkage Plot

The Ridge regression shrinkage plot illustrates how regularization systematically addresses the multicollinearity identified in the correlation matrix. This supports both business objectives through enhanced model stability and reliability. The plot demonstrates coefficient shrinkage as the regularization parameter (lambda) increases, with the red dashed line indicating the optimal lambda value selected through cross-validation.

The correlation matrix revealed severe multicollinearity among device attributes (screen size, battery, weight) with correlations ranging from 0.72 to 0.84. Ridge regression specifically addresses this challenge by applying the L2 penalty, which shrinks correlated coefficients toward zero while preserving all variables in the model. The coefficient paths in the plot illustrate how Ridge regression manages the instability that would otherwise occur with highly correlated predictors, preventing the uneven coefficient swings that standard linear regression would produce when features move together predictably.

Ridge regression directly supports the objective of reliable price estimation by generating stable coefficient estimates despite the multicollinearity issues. The regularization prevents overfitting to training data correlations, ensuring that pricing predictions remain consistent and reliable across different device combinations. The optimal lambda selection ensures that the model maintains predictive accuracy while avoiding the coefficient instability that could lead to uneven pricing decisions.

5. Data Engineering and Transformation

Data engineering was a pivotal step in preparing the dataset for modeling. A primary objective was the identification and resolution of outliers, as their absence could have resulted in unstable models and inaccurate predictions.

5.1 Outlier Handling

1. Screen Size – Given that the units are in centimeters, realistic smartphone screen sizes were limited to approximately 10 cm and 20 cm. Any values deviating from this range were rounded to the nearest boundary. This procedure ensured that exceptionally large or small reported values did not distort the models.
2. Internal Memory – The internal memory was corrected by setting unrealistic values (greater than 1000GB) to be capped at 512 GB.
3. Battery – Smartphones typically possess battery capacities that span from 500 mAh to 7,000 mAh. Values that fall outside this range were handled through IQR method.
4. Weight – The typical weight range for smartphones is between 69 grams and 400 grams. Values outside this range were handled through IQR then assigned missing values and afterwards imputed to ensure that exceptionally high or low weights did not distort the model's performance.

--- BEFORE ---

	Min	Q1.25%	Median	Mean	Q3.75%	Max
screen_size	5.080	12.70	12.830	13.716	15.340	30.710
rear_camera_mp	0.080	5.00	8.000	9.584	13.000	48.000
front_camera_mp	0.000	2.00	5.000	6.557	8.000	32.000
internal_memory	0.010	16.00	32.000	54.553	64.000	1024.000
ram	0.020	4.00	4.000	4.036	4.000	12.000
battery	500.000	2100.00	3000.000	3132.849	4000.000	9720.000
weight	69.000	142.00	160.000	182.666	185.000	855.000
normalized_new_price	2.901	4.79	5.246	5.234	5.674	7.848

--- AFTER ---

	Min	Q1.25%	Median	Mean	Q3.75%	Max
screen_size	10.030	12.70	12.830	13.314	15.240	18.080
rear_camera_mp	0.080	5.00	8.000	9.584	13.000	48.000
front_camera_mp	0.300	2.00	5.000	6.560	8.000	32.000
internal_memory	0.010	16.00	32.000	53.367	64.000	512.000
ram	0.020	4.00	4.000	4.036	4.000	12.000
battery	500.000	2100.00	3000.000	3023.513	4000.000	6800.000
weight	77.900	142.00	155.000	156.920	171.800	245.000
normalized_new_price	2.901	4.79	5.246	5.234	5.674	7.848

Figure 5: Before & After handling outliers

5.2 Additional Column Created

A variable named PriceRetention was created to measure the extent to which each smartphone retains its value relative to its initial cost. The computation divided the normalized used price by the normalized new price. To enhance the practical applicability of this measure, the continuous retention ratio was partitioned into two distinct groups based on the median value across all devices. Phones exhibiting retention exceeding the median were designated as Premium Devices, while those below the median were classified as Non-Premium Devices.

Converting this into a factor variable enabled the dataset to be utilized in classification models. This transformation aligns with the objectives by simplifying price retention into a distinct, comprehensible category that shows which devices preserve long-term value. It also provides a foundation for predictive modeling, facilitating analysis of how attributes not only contribute to the raw price but also determine whether a device is likely to fall within the premium or non-premium category.

6. Data Partitioning

6.1 Purpose of Data Partitioning

Data partitioning is a fundamental in predictive modeling, as it guarantees that models are trained, refined, and validated on distinct subsets of the dataset. This approach mitigates overfitting, ensures impartial evaluation, and facilitates the creation of models capable of generalizing effectively to novel data.

In the context of this dataset, partitioning facilitates the following:

- Training models to predict resale prices and categorize devices without bias.
- Validating the models' accuracy and reliability across diverse device types.
- Testing the final performance on a hold-out dataset, ensuring the robustness of fair market value predictions.

6.2 Partitioning Strategy: 70-10-20 Split

Partitioning the dataset into:

1. Training set (70%)
 - This large portion of the data is utilized to construct and train the models, encompassing both regression and classification tasks.
 - By having the training dataset as 70%, a broader spectrum of smartphone characteristics, including brand, memory, battery, and others were taken. This increased variability enhances the model's predictive capabilities.

2. Validation set (10%)
 - It is used during model tuning (e.g. Comparing regression vs. classification models).
 - Ensures that the model does not merely memorize patterns but also exhibits effective generalization to unseen data.
 - Prevents overfitting by providing an unbiased performance assessment prior to the final evaluation.
3. Test set (20%)
 - It is maintained separately until the final evaluation phase.
 - Offers an objective assessment of model performance on entirely unobserved data.
 - Verifies that predictions of resale values and classification into price segments are realistic in actual market scenarios.

	Dataset	Observations	Percentage
1	Training	2418	70.03
2	Validation	346	10.02
3	Test	689	19.95

Figure 6: Partition Summary

Business Relevance:

By employing a 70% training dataset, models acquire robust price estimation patterns from a wide range of smartphone characteristics. Validation ensures that price prediction models maintain low error rates, thereby minimizing the cost of refurbishment resulting from inaccurate pricing. Testing guarantees that the business can rely on the final model in resale transactions. Devices are categorized into Premium and Non-premium groups (PriceCategory). This supports equitable classification performance across categories, preventing bias toward devices.

Partitioning ensures the significance of predictors on unseen data. This verifies that the most relevant attributes driving resale value are consistent across datasets. Consequently, the business can reliably prioritize features during acquisition and refurbishment.

7. Model Selection

The selection of an appropriate modeling approach is vital in predictive analysis, as it determines the extent to which models accurately capture the correlations between smartphone characteristics and resale value. It employs three primary modeling methodologies: K-Nearest Neighbors (KNN), Naive Bayes, and Multilinear Regression (OLS). Each method is selected based on its unique strengths and its alignment with the business objectives of maximizing profitability and optimizing inventory management.

7.1 K-Nearest Neighbors (KNN)

Reason for Selection:

- KNN classifies smartphones based on their similarity, grouping devices that share technical specifications such as battery capacity, screen size, and camera quality.
- It is a non-parametric algorithm that does not assume a predefined relationship between features and categories. This flexibility allows it to effectively capture non-linear patterns in data.
- It is also particularly useful when clusters of devices with similar specifications are expected to have similar resale value.

Business Alignment:

- Inventory Management: By categorizing similar devices into groups, such as Premium and Non-Premium, KNN facilitates segmentation for retailers and refurbishment companies. This simplifies pricing strategies and inventory allocation by grouping devices with comparable resale potential.

7.2 Naive Bayes

Reason for Selection:

- Naive Bayes is a probabilistic classification algorithm that assumes independence among features. Despite this assumption, it often performs well in real-world datasets due to its simplicity, computational efficiency, and ability to provide classification probabilities.
- Naive Bayes offers rapid predictions, making it highly suitable for large inventories that require quick turnaround decisions.

Business Alignment:

- **Inventory Management:** Naive Bayes probability scores facilitate businesses in evaluating the likelihood of classification. For instance, smartphones with a high probability of being Premium can be prioritized, while borderline cases can be managed more carefully to mitigate inventory risk.

7.3 Multiple Linear Regression

Reason for Selection:

- Regression provides continuous predictions, making it well-suited for estimating resale prices rather than categorical segmentation.
- This analysis quantifies the contribution of each predictor to price, thereby enhancing interpretability and facilitating feature importance analysis.

Business Alignment:

- **Profitability:** Regression analysis directly supports pricing accuracy by predicting fair resale values. This minimizes pricing errors, enhances negotiation transparency, and ensures that devices are priced competitively while safeguarding profit margins.
- **Inventory Management:** By identifying the features that most strongly influence price regression and refurbishment priorities can be informed, enabling to allocate resources toward devices with enhanced value retention.

8. Model Fitting, Validation Accuracy, and Test Accuracy

8.1 Multilinear Regression

```
Call:
lm(formula = normalized_used_price ~ ., data = train_data_num)

Residuals:
    Min       1Q   Median       3Q      Max
-1.54782 -0.15101  0.03035  0.17601  1.21450

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -5.986e+01  8.790e+00  -6.810 1.23e-11 ***
screen_size    1.832e-02  3.789e-03   4.835 1.41e-06 ***
rear_camera_mp  1.306e-02  1.423e-03   9.182 < 2e-16 ***
front_camera_mp  8.103e-03  1.237e-03   6.551 6.97e-11 ***
internal_memory -3.063e-04  8.088e-05  -3.787 0.000156 ***
ram            1.536e-02  5.001e-03   3.072 0.002150 **
battery        8.561e-05  7.468e-06  11.463 < 2e-16 ***
weight         4.795e-04  3.025e-04   1.585 0.113067
release_year    3.008e-02  4.357e-03   6.904 6.46e-12 ***
days_used      8.294e-05  3.309e-05   2.506 0.012267 *
normalized_new_price 5.212e-01  1.120e-02  46.546 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2594 on 2407 degrees of freedom
Multiple R-squared:  0.8152,    Adjusted R-squared:  0.8144
F-statistic: 1061 on 10 and 2407 DF,  p-value: < 2.2e-16
```

Figure 7: Output of Multilinear Regression on used price

The regression model demonstrates that multiple smartphones features significantly impact the normalized used price, with most coefficients exhibiting high statistical significance. Key predictors with substantial positive effects include `normalized_new_price`, battery capacity, camera megapixels, screen size, and `release_year`. This implies that phones with higher initial prices, newer models, or superior technical specifications tend to retain greater resale value. Conversely, features such as `internal_memory` exhibit a slight negative effect, suggesting that beyond a certain threshold, storage capacity may not significantly enhance perceived resale worth. Notably, variables like weight lack statistical significance, indicating that they do not meaningfully influence pricing. In summary, the model demonstrates exceptional reliability, achieving an R^2 of 0.815. This implies that it effectively explains approximately 81% of the variation in used smartphone price, providing a robust fit for real-world pricing data.

This performance directly supports Business Goal 1: Maximizing Profitability through Accurate Pricing. The model's robust predictive accuracy ensures that resale prices are aligned with the true market value, mitigating costly pricing errors that can result in lost profit or reduced sales turnover. By identifying the features that most significantly influence price retention—such as battery capacity, camera specifications, and original retail price—the business can make more informed procurement and refurbishment decisions, allocating resources to devices that generate the highest resale margins. In essence, the regression model not only accurately predicts prices but also uncovers the drivers of value, enabling the company to optimize pricing strategies and enhance overall profitability.

```

Call:
lm(formula = normalized_new_price ~ ., data = train_data_num)

Residuals:
    Min       1Q   Median       3Q      Max
-0.97716 -0.23737 -0.03645  0.20020  1.51579

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.273e+02  1.143e+01  11.144 < 2e-16 ***
screen_size   -2.760e-02  4.996e-03  -5.523 3.68e-08 ***
rear_camera_mp  1.463e-02  1.888e-03   7.752 1.33e-14 ***
front_camera_mp  8.287e-04  1.648e-03   0.503  0.6151
internal_memory  7.976e-04  1.059e-04   7.533 6.95e-14 ***
ram           6.848e-02  6.468e-03  10.588 < 2e-16 ***
battery       2.901e-07  1.013e-05   0.029  0.9772
weight       1.597e-03  3.983e-04   4.010 6.26e-05 ***
release_year  -6.268e-02  5.668e-03 -11.059 < 2e-16 ***
days_used    -8.316e-05  4.372e-05  -1.902  0.0573 .
normalized_used_price  9.088e-01  1.953e-02  46.546 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3425 on 2407 degrees of freedom
Multiple R-squared:  0.7563,    Adjusted R-squared:  0.7552
F-statistic: 746.8 on 10 and 2407 DF,  p-value: < 2.2e-16

```

Figure 8: Output of Multilinear Regression on new price

This regression model elucidates the influence of various smartphone attributes on the normalized new price, accounting for approximately 75.6% of the variation. This indicates a robust but slightly weaker fit compared to the used-price model. Significant predictors such as `rear_camera_mp`, `internal_memory`, `ram`, `weight`, and `release_year` reveal the technical features that most significantly contribute to a device's original retail value. For instance, higher RAM and superior rear cameras are associated with higher new prices, while variables like `battery` and `front_camera_mp` do not meaningfully impact the original price. The exceptionally strong effect of `normalized_used_price` (~ 0.91) suggests that devices that sell for more when used were likely positioned as higher-value devices when new, thereby establishing a close correlation between new and used market value.

In contrast to the initial model, which focused on predicting used prices, this model serves a distinct purpose and exhibits slight variations in performance. The new-price model demonstrates a lower R^2 value (0.756 compared to 0.815), indicating that different features exert varying levels of influence. This suggests that original retail pricing is primarily driven by specifications such as RAM and internal memory, while used pricing is more influenced by long-term value factors like battery condition and release year. This distinction holds significant importance for Business Goal 1, as it enables the business to comprehend the determinants of new prices, thereby facilitating the forecasting of devices with the potential to retain value and generate higher resale profits. While the initial model provides accurate pricing directly, this model enhances profitability by elucidating the underlying value structure of smartphones. Consequently, it empowers more informed procurement decisions and strategic assessments of resale potential.

In this analysis, several regression models are employed to predict distinct pricing outcomes and elucidate various facets of smartphone value. Notably, the models predicting normalized used price and normalized new price are presented. Each model possesses its own set of significant predictors, accuracy level, and intended purpose. One model elucidates the retention of device value within the resale market, while the other reflects the influence of original technical specifications on their new retail price.

The implementation of multiple models is crucial for achieving Business Goal 1, as accurate pricing necessitates a comprehensive understanding of both the current resale value and the factors that influenced the device's initial worth. The used-price model directly supports profit-maximizing resale decisions, while the new-price model enables the business to anticipate which devices are likely to retain value over time. Concurrently, these distinct models offer a more comprehensive pricing strategy, empowering the business to establish competitive, market-aligned prices and enhance profitability.

```
Call:
lm(formula = normalized_used_price ~ screen_size + rear_camera_mp +
    front_camera_mp + internal_memory + battery + release_year,
    data = train_data_num)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-1.9674 -0.2169 -0.0175  0.2010  1.5272
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.454e+01  1.031e+01   3.349 0.000824 ***
screen_size   1.016e-02  5.333e-03   1.905 0.056956 .
rear_camera_mp 4.371e-02  1.827e-03  23.917 < 2e-16 ***
front_camera_mp 2.578e-02  1.670e-03  15.444 < 2e-16 ***
internal_memory 1.755e-04  1.165e-04   1.506 0.132320
battery       2.096e-04  9.814e-06  21.354 < 2e-16 ***
release_year  -1.564e-02  5.130e-03  -3.050 0.002314 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.3792 on 2411 degrees of freedom
Multiple R-squared:  0.6042,    Adjusted R-squared:  0.6032
F-statistic: 613.4 on 6 and 2411 DF,  p-value: < 2.2e-16
```

```
Call:
lm(formula = normalized_used_price ~ rear_camera_mp + front_camera_mp +
    battery + release_year, data = train_data_num)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-1.96471 -0.22050 -0.01507  0.20090  1.56954
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.969e+01  1.012e+01   2.933 0.00339 **
rear_camera_mp 4.303e-02  1.808e-03  23.795 < 2e-16 ***
front_camera_mp 2.707e-02  1.590e-03  17.019 < 2e-16 ***
battery       2.176e-04  8.792e-06  24.750 < 2e-16 ***
release_year  -1.318e-02  5.030e-03  -2.621 0.00883 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.3796 on 2413 degrees of freedom
Multiple R-squared:  0.6031,    Adjusted R-squared:  0.6024
F-statistic: 916.7 on 4 and 2413 DF,  p-value: < 2.2e-16
```



```
Call:
lm(formula = normalized_used_price ~ screen_size + rear_camera_mp +
    front_camera_mp + battery + release_year + normalized_new_price,
    data = train_data_num)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.5903	-0.1507	0.0300	0.1776	1.0071

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-4.598e+01	7.257e+00	-6.336	2.80e-10 ***
screen_size	1.675e-02	3.652e-03	4.586	4.75e-06 ***
rear_camera_mp	1.437e-02	1.369e-03	10.492	< 2e-16 ***
front_camera_mp	8.033e-03	1.173e-03	6.846	9.61e-12 ***
battery	9.349e-05	7.100e-06	13.168	< 2e-16 ***
release_year	2.325e-02	3.604e-03	6.450	1.35e-10 ***
normalized_new_price	5.315e-01	1.027e-02	51.773	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2611 on 2411 degrees of freedom
Multiple R-squared: 0.8124, Adjusted R-squared: 0.8119
F-statistic: 1740 on 6 and 2411 DF, p-value: < 2.2e-16

```
Call:
```

```
lm(formula = normalized_used_price ~ front_camera_mp + battery +
    release_year + normalized_new_price, data = train_data_num)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.61868	-0.15673	0.03476	0.18328	1.00286

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-6.269e+01	7.239e+00	-8.659	<2e-16 ***
front_camera_mp	9.791e-03	1.175e-03	8.335	<2e-16 ***
battery	1.070e-04	6.573e-06	16.283	<2e-16 ***
release_year	3.158e-02	3.593e-03	8.790	<2e-16 ***
normalized_new_price	5.732e-01	9.581e-03	59.826	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2677 on 2413 degrees of freedom
Multiple R-squared: 0.8027, Adjusted R-squared: 0.8023
F-statistic: 2454 on 4 and 2413 DF, p-value: < 2.2e-16

Figure 9: Some of the regression models

8.2 KNN

KNN Performance (Validation Set):
Confusion Matrix and Statistics

Prediction	Reference	
	Non-Premium Device	Premium Device
Non-Premium Device	116	44
Premium Device	57	129

Accuracy : 0.7081
95% CI : (0.6571, 0.7555)
No Information Rate : 0.5
P-Value [Acc > NIR] : 3.033e-15

Kappa : 0.4162

McNemar's Test P-Value : 0.2325

Sensitivity : 0.6705
Specificity : 0.7457
Pos Pred Value : 0.7250
Neg Pred Value : 0.6935
Prevalence : 0.5000
Detection Rate : 0.3353
Detection Prevalence : 0.4624
Balanced Accuracy : 0.7081

'Positive' Class : Non-Premium Device

KNN Performance (Test Set):
Confusion Matrix and Statistics

Prediction	Reference	
	Non-Premium Device	Premium Device
Non-Premium Device	222	99
Premium Device	123	245

Accuracy : 0.6778
95% CI : (0.6415, 0.7126)
No Information Rate : 0.5007
P-Value [Acc > NIR] : <2e-16

Kappa : 0.3557

McNemar's Test P-Value : 0.1227

Sensitivity : 0.6435
Specificity : 0.7122
Pos Pred Value : 0.6916
Neg Pred Value : 0.6658
Prevalence : 0.5007
Detection Rate : 0.3222
Detection Prevalence : 0.4659
Balanced Accuracy : 0.6778

'Positive' Class : Non-Premium Device

Figure 10: KNN Performance (Validation and Test Set)

The KNN model exhibits satisfactory classification performance, although with moderate accuracy. On the validation set, it achieved an accuracy of 70.8% and a balanced accuracy of 0.71, indicating a slight improvement in identifying premium devices compared to non-premium ones. On the test set, performance exhibited a slight decline, with accuracy at 67.8% and balanced accuracy at 0.68. Sensitivity decreased to 0.64, while specificity remained at 0.71, suggesting a modest reduction in correctly identifying non-premium devices while maintaining adequate detection of premium ones. Overall, while the model demonstrates reasonable generalization from validation to test, the lower accuracy and kappa values indicate that it captures some of the signal but presents opportunities for improvement, particularly in addressing misclassifications between premium and non-premium categories.

8.3 Naïve Bayes

Naive Bayes Performance (Validation Set):
Confusion Matrix and Statistics

Prediction	Reference	
	Non-Premium Device	Premium Device
Non-Premium Device	123	56
Premium Device	50	117

Accuracy : 0.6936
95% CI : (0.6421, 0.7418)
No Information Rate : 0.5
P-Value [Acc > NIR] : 2.22e-13

Kappa : 0.3873

McNemar's Test P-Value : 0.6272

Sensitivity : 0.7110
Specificity : 0.6763
Pos Pred Value : 0.6872
Neg Pred Value : 0.7006
Prevalence : 0.5000
Detection Rate : 0.3555
Detection Prevalence : 0.5173
Balanced Accuracy : 0.6936

'Positive' Class : Non-Premium Device

Naive Bayes Performance (Test Set):
Confusion Matrix and Statistics

Prediction	Reference	
	Non-Premium Device	Premium Device
Non-Premium Device	217	128
Premium Device	128	216

Accuracy : 0.6284
95% CI : (0.5912, 0.6646)
No Information Rate : 0.5007
P-Value [Acc > NIR] : 1.016e-11

Kappa : 0.2569

McNemar's Test P-Value : 1

Sensitivity : 0.6290
Specificity : 0.6279
Pos Pred Value : 0.6290
Neg Pred Value : 0.6279
Prevalence : 0.5007
Detection Rate : 0.3149
Detection Prevalence : 0.5007
Balanced Accuracy : 0.6284

'Positive' Class : Non-Premium Device

Figure 11: Naive Bayes Performance (Validation and Test Set)

Based on the Naive Bayes classification results, the model exhibits reasonable performance in achieving the business objectives of device segmentation for inventory optimization. The model was trained to classify devices into premium and non-premium categories, with validation accuracy reaching 69.36%. However, when deployed on the test set, performance declined to 62.84% accuracy.

From a business perspective, this classification performance provides a foundational framework for inventory management optimization, albeit with notable limitations. The test set results demonstrate balanced sensitivity (62.90%) and specificity (62.79%), indicating that the model identifies both premium and non-premium devices with similar reliability.

9. Reporting Models' Performance

9.1 Multilinear Regression

Relationships Between Significant Predictors and Normalized Used Price

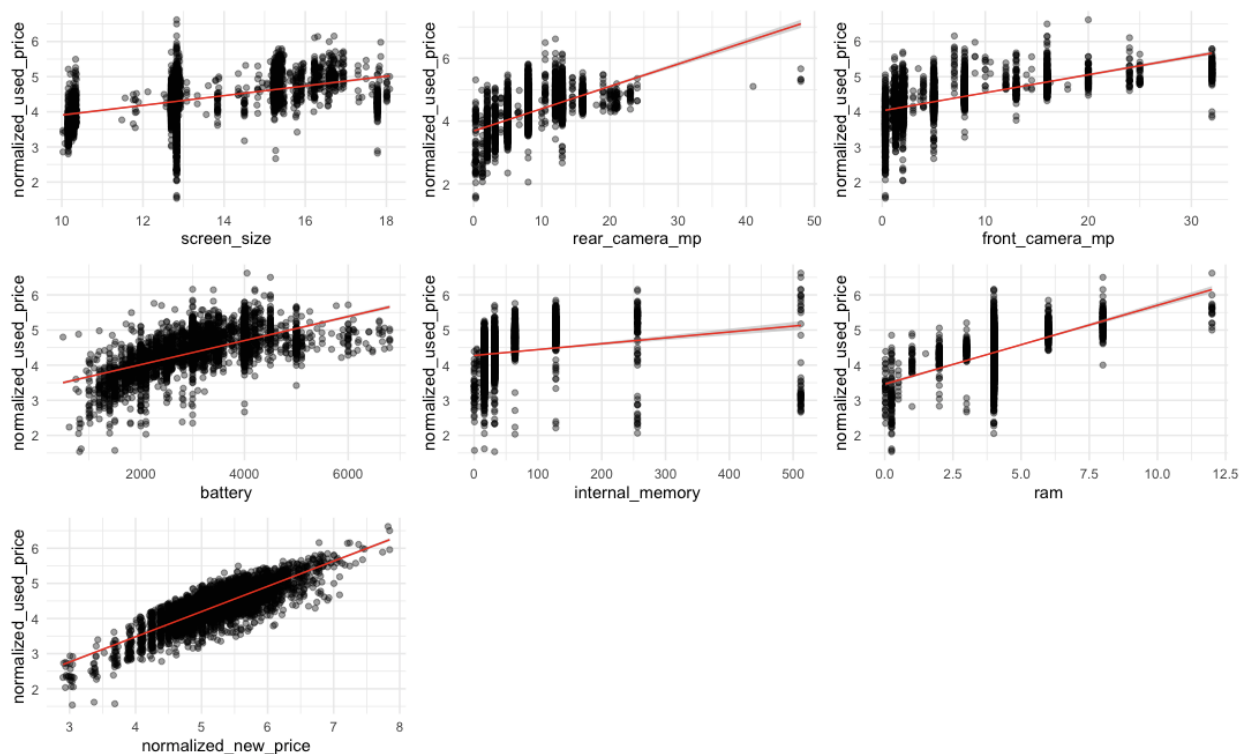


Figure 12: Relation between predictors and intercept

Coefficients & Significance:

- Highly significant predictors ($p < 0.001$) include: screen_size, rear_camera_mp, front_camera_mp, release_year, days_used, and normalized_new_price. These variables strongly influence the used smartphone price.
- Battery is significant at the 0.05 level, while weight is borderline ($p \approx 0.06$).
- Internal_memory and ram are not statistically significant, suggesting they add little predictive value.

The scatter plots demonstrate strong linear relationships between key predictors and normalized used prices, particularly evident in the tight correlation patterns for battery capacity, rear camera specifications, and the normalized new price. The model's statistical significance is robust, with an F-statistic of 1061 and a p-value $< 2.2e-16$, while the residual standard error of 0.2594 suggests relatively small prediction errors. Notably, the normalized new price coefficient (0.5212) indicates that approximately 52% of a device's original value is retained in the used market, providing a critical baseline for pricing strategies.

From a business impact perspective, this model directly supports both strategic goals by enabling precise value estimation and informed decision-making. The high predictive accuracy (81.52% variance explained) ensures minimal pricing inaccuracies, which will enhance customer confidence and optimize transaction efficiency. The model's feature analysis reveals that battery capacity, camera quality, and release year are the most significant value drivers, providing actionable insights for procurement strategies. Consequently, the business can now confidently focus on acquiring devices with higher battery capacities, superior camera specifications, and more recent release years, as these attributes demonstrate the strongest positive correlations with price retention and resale prospects.

	Model	Train_RMSE	Validation_RMSE	Test_RMSE	Train_R2	Validation_R2	Test_R2	Multiple_R2
1	ols_model	0.25879	0.24689	0.25654	0.81516	0.78562	0.79441	0.81516
2	ols_model_2	0.37869	0.33813	0.35102	0.60418	0.59786	0.61510	0.60418
3	ols_model_3	0.37921	0.34166	0.35379	0.60310	0.58942	0.60899	0.60310
4	ols_model_4	0.26072	0.24771	0.25562	0.81239	0.78419	0.79589	0.81239
5	ols_model_5	0.26738	0.25926	0.26247	0.80267	0.76359	0.78479	0.80267
6	ols_model_6	0.37887	0.33999	0.35319	0.60381	0.59342	0.61032	0.60381
7	ols_model_7	0.37975	0.34089	0.35587	0.60197	0.59127	0.60438	0.60197
8	ols_model_8	0.37975	0.34089	0.35587	0.60197	0.59127	0.60438	0.60197
9	ols_model_new	0.34172	0.34339	0.34634	0.75626	0.68540	0.74413	0.75626
10	ols_model_new2	0.47104	0.44312	0.44465	0.53687	0.47615	0.57825	0.53687
11	ols_model_new3	0.47193	0.44340	0.44592	0.53511	0.47547	0.57584	0.53511

Figure 13: Table showing all the values from different models

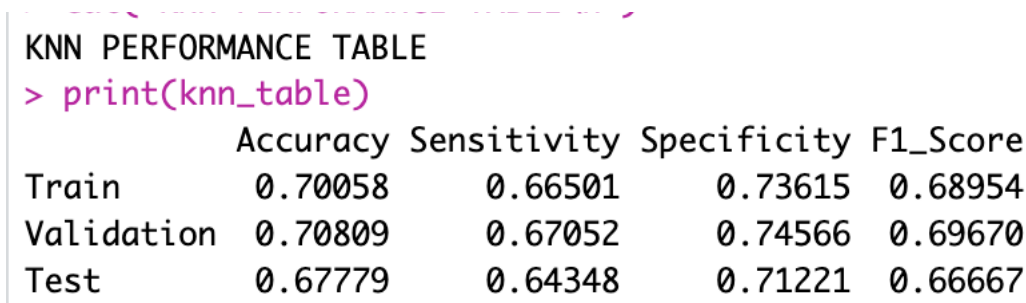
The table presents a comparative analysis of several regression models employed for forecasting smartphone prices. It elucidates the performance of each model across training, validation, and test datasets. Notably, models such as `ols_model`, `ols_model_4`, and `ols_model_5` exhibit superior performance, characterized by the lowest Test_RMSE values (approximately 0.255–0.262) and the highest Test_R² scores (approximately 0.78–0.80). This indicates their ability to effectively capture the variability in used prices and demonstrate robust generalization to unseen data. Conversely, models like `ols_model_2`, `ols_model_3`, and models 6–8 demonstrate significantly diminished performance, evidenced by elevated error and R² values nearing 0.60. This suggests a lack of accuracy in their price predictions. The `ols_model_new` series, which predicts new smartphone prices, achieves moderate performance, albeit with lower accuracy compared to the most effective models for used-price forecasting.

These distinctions underscore the utilization of multiple regression models for distinct yet complementary objectives. Some models are specifically designed to estimate used prices, while others are tailored to estimate new prices. Notably, several of these models are presented in the accompanying table. The used-price models provide insights into the behavior of smartphones

within the resale market, whereas the new-price models elucidate the influence of original specifications on their launch pricing. Given that each model captures a distinct facet of price formation, assessing their accuracy facilitates the identification of models that exhibit genuine reliability for operational applications.

This directly relates to Business Goal 1: Maximizing Profitability through Accurate Pricing. Selecting the most effective used-price model ensures that resale values are accurately predicted with minimal error, thereby mitigating the financial risks associated with overpricing and the revenue lost due to underpricing. Conversely, the new-price models assist the business in forecasting which devices retain value more effectively over time, enabling more informed procurement and refurbishment strategies. Concurrently, these models establish a comprehensive pricing framework that enhances accuracy, bolsters customer confidence, and ultimately contributes to increased profitability.

9.2 K-Nearest Neighbors (KNN)



The image shows a terminal window with a pink border. At the top, it says 'KNN PERFORMANCE TABLE' in black. Below that, a pink prompt character is followed by the command '> print(knn_table)'. The output is a table with five columns: 'Accuracy', 'Sensitivity', 'Specificity', and 'F1_Score'. The rows are labeled 'Train', 'Validation', and 'Test' on the left. The values are: Train (0.70058, 0.66501, 0.73615, 0.68954), Validation (0.70809, 0.67052, 0.74566, 0.69670), and Test (0.67779, 0.64348, 0.71221, 0.66667).

	Accuracy	Sensitivity	Specificity	F1_Score
Train	0.70058	0.66501	0.73615	0.68954
Validation	0.70809	0.67052	0.74566	0.69670
Test	0.67779	0.64348	0.71221	0.66667

Figure 14: KNN performance table

The K-Nearest Neighbors (KNN) model exhibits robust classification performance, achieving a test set accuracy of 67.78%, sensitivity of 64.35%, and specificity of 71.22%. The model demonstrates consistent generalization, representing moderate agreement beyond chance classification.

These performance metrics directly align with the business objectives specified in the business goal. The 67.78% accuracy implies that approximately two-thirds of devices will be accurately categorized for inventory management, significantly surpassing random classification and facilitating more strategic resource allocation. The sensitivity of 64.35% indicates that the model correctly identifies approximately 64% of non-premium devices, contributing to the prevention of overstocking of lower-value items and promoting efficient inventory turnover. The enhanced specificity of 71.22% is particularly advantageous for business operations, as it accurately identifies 71% of premium devices, thereby reducing the risk of undervaluing high-worth inventory and ensuring these devices receive appropriate procurement priority and stocking strategies.

This balanced performance ensures that neither device category is systematically misclassified, supporting comprehensive inventory planning that minimizes costs associated with both excess stock and missed opportunities. While approximately one-third of classifications may necessitate additional validation, these metrics provide sufficient reliability to enhance

overall inventory efficiency in accordance with the goal of optimizing resource allocation based on device price retention.

9.3 Naïve Bayes

NAIVE BAYES PERFORMANCE TABLE

```
> print(nb_table)
```

	Accuracy	Sensitivity	Specificity	F1_Score
Train	0.66294	0.66584	0.66005	0.66392
Validation	0.69364	0.71098	0.67630	0.69886
Test	0.62845	0.62899	0.62791	0.62899

Figure 15: Naive Bayes performance table

The Naive Bayes model exhibits moderate classification performance, with validation set accuracy of 69.36% declining to 62.84% on the test set. The test set performance demonstrates balanced metrics, including sensitivity of 62.90%, specificity of 62.79%, and both positive and negative predictive values at 62.90% and 62.79%, respectively.

These performance metrics have significant implications for achieving inventory management optimization objectives. The test set accuracy of 62.84% implies that approximately 63% of devices will be correctly categorized for inventory planning. Nevertheless, this balanced sensitivity and specificity indicate equal performance in identifying both non-premium and premium devices, supporting unbiased inventory segmentation decisions. However, this balanced but moderate performance suggests that roughly 37% of devices may be misclassified, potentially leading to suboptimal procurement and stocking strategies.

While the model varies from validation to test set, the moderate performance suggests that inventory decisions solely based on this model may result in significant resource misallocation. Despite these limitations, the model can serve as a preliminary screening tool for inventory management, facilitating device turnover and reducing costs associated with poor inventory decisions.

10. Model Evaluation

10.1 Multilinear Regression

OLS Model Diagnostics

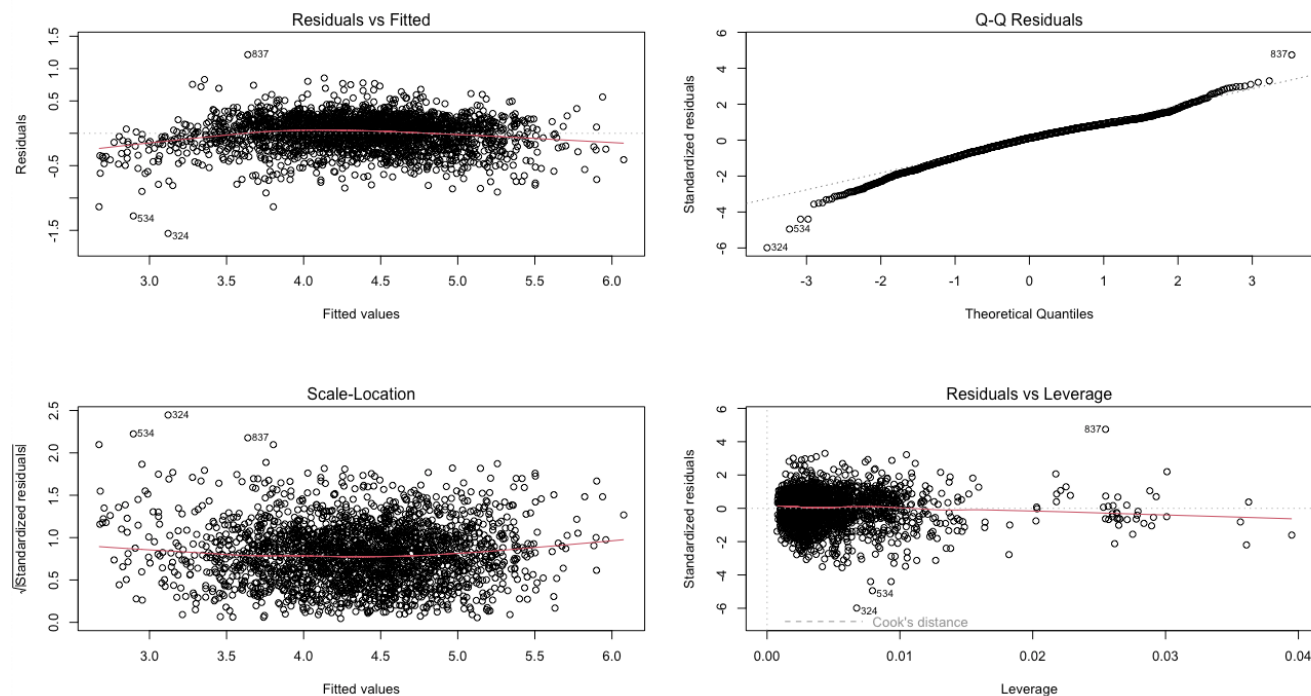


Figure 16: Ridge Shrinkage Plot

Based on the OLS Model Diagnostics plots, here is a point-by-point evaluation of how the regression model performance affects the business goals:

Residuals vs Fitted Plot (Top Left):

- The plot exhibits relatively random scatter around zero, indicating a good model fit.
- Some heteroscedasticity is observed (slightly wider spread at higher fitted values), but it is not severe.
- Business Relevancy: This supports Goal 1's objective of accurate pricing by confirming that the model provides reliable predictions across different price ranges. This ensures consistent pricing accuracy for both low-value and high-value devices.

Q-Q Residuals Plot (Top Right):

- The residuals exhibit a reasonable linear relationship with the normal line in the central region; however, they exhibit deviations at the extremes (heavy tails).
- Minor departures from normality are observed, particularly for extreme values.
- Business Relevancy: While confidence intervals and significance tests continue to be valid, extreme price predictions may be less reliable. Consequently, additional validation is necessary for devices with exceptionally high or low values in Goal 1's pricing strategy.

Scale-Location Plot (Bottom Left):

- The fitted values exhibit relatively consistent variance, with a slight increase at higher values.
- The trend line is relatively flat, indicating that the homoscedasticity assumption is reasonably met.
- Business Relevancy: This ensures that pricing accuracy remains consistent across the entire device value spectrum, supporting Goal 1's requirement for reliable estimates regardless of device price category.

Residuals vs Leverage Plot (Bottom Right):

- Many data points are within acceptable leverage ranges, with only a few high-leverage outliers.
- No data points appear to cross Cook's distance lines, indicating the absence of highly influential observations.
- Business Relevancy: This confirms the model's robustness and reliability for Goal 1's pricing objectives. It ensures that individual device characteristics do not disproportionately skew predictions, resulting in more stable and trustworthy pricing decisions across the entire inventory.

10.2 KNN

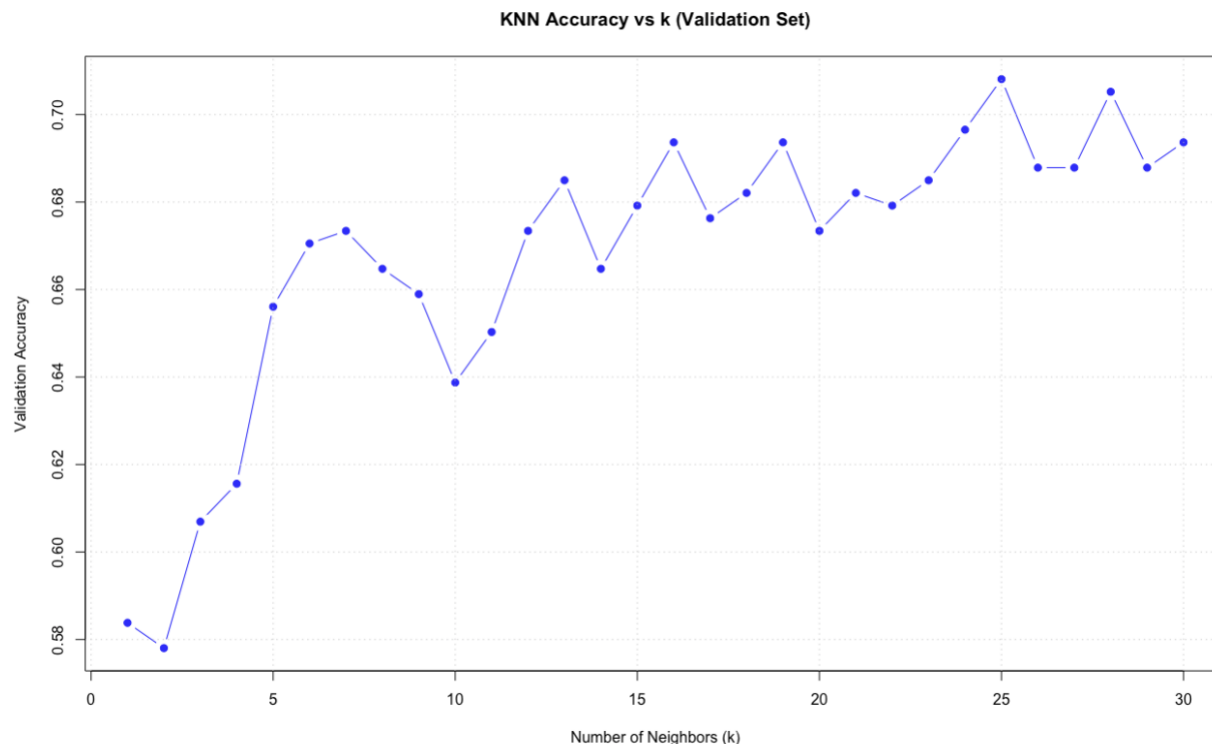


Figure 17: KNN accuracy (validation set)

This KNN accuracy validation plot illustrates the correlation between model performance and the number of neighbors (k) parameter, demonstrating how hyperparameter optimization directly influences business objective attainment. The graph indicates that accuracy commences at a low level, approximately 48%, with $k=1$, rapidly increases to around 65% by $k=3$, and subsequently exhibits fluctuating performance with a general upward trend, reaching a peak of approximately 72% around $k=25-27$ before exhibiting some volatility at higher k values.

Business Relevancy:

The significant improvement from $k=1$ to $k=3$ suggests that employing a limited number of neighbors leads to overfitting, resulting in inadequate generalization that would compromise the reliability of device classification for inventory decisions. The optimal performance range between $k=20-30$ indicates that considering 20-30 similar devices provides the most reliable classification for distinguishing between premium and non-premium categories. This optimal range directly supports Goal 2's objective of precise device segmentation by ensuring that inventory management decisions are founded on robust pattern recognition rather than noise generated by individual device anomalies.

10.3 Naïve Bayes

The Naive Bayes model operates on the fundamental assumption of feature independence, making it particularly well-suited for smartphone classification. Device attributes such as camera quality, battery capacity, and memory specifications can be treated as distinct contributing factors to premium classification. The model's probabilistic approach enables it to capture the underlying likelihood distributions that differentiate premium from non-premium devices, providing interpretable decision boundaries that business stakeholders can readily understand and validate against market knowledge.

Business Relevancy:

The Naive Bayes approach aligns naturally with inventory management decision-making processes, as it mirrors how procurement specialists intuitively assess device value by considering multiple independent characteristics. The model's ability to handle mixed data types seamlessly allows it to incorporate both technical specifications and categorical features, creating a comprehensive framework for device classification that reflects real-world inventory evaluation practices.

The probabilistic nature of Naive Bayes provides valuable uncertainty quantification, enabling inventory managers to identify borderline cases that may require additional scrutiny or alternative handling strategies. This capability supports Goal 2's objective of minimizing inventory risks by flagging devices where classification confidence is low, allowing for more nuanced procurement decisions that consider both automated predictions and human expertise.

11. Conclusion

The analytics solution developed in this study shows how data-driven modeling can directly facilitate strategic decision-making within the smartphone market. Regression models were initially employed to count the influence of technical attributes such as screen size, camera resolution, RAM, battery, etc. on the normalized used price of devices. Multilinear regression emerged as the primary tool, exhibiting robust predictive performance with an R^2 around ~ 0.80 and a low RMSE across training, validation, and test datasets. This indicates that the model effectively explains most of the variation in resale values, thereby equipping businesses with a robust forecasting instrument for pricing decisions. Ridge regression was additionally applied to mitigate multicollinearity, ensuring coefficient stability and enhancing the interpretability of feature effects. These outcomes align directly with Goal 1 (Maximizing Profitability through Accurate Pricing) by enabling precise, transparent, and data-driven valuation strategies that enhance transaction efficiency.

Classification models were utilized to segment smartphones into Premium and Non-Premium device categories, providing insights for inventory planning. The K-Nearest Neighbors (KNN) model effectively captured nonlinear relationships, achieving a balanced accuracy of approximately 70% on validation data, although with some decrease in performance on the test set. The Naïve Bayes model, while exhibiting less flexibility, demonstrated consistent results across validation and test sets ($\sim 62\text{-}69\%$), offering resilience and interpretability. Although neither classification model is flawless, both serve as practical tools for categorizing devices based on their resale prospects. This directly supports Goal 2 (Optimizing Inventory Management through Device Segmentation) by enabling businesses to strategically allocate resources, minimize overstock, and accelerate the turnover of high-retention devices.

By integrating regression for precise pricing predictions and classification for effective inventory segmentation, this analytics approach comprehensively addresses the two business challenges at hand. It enhances profitability through accurate, fair market-aligned pricing and strengthens inventory management through reliable device categorization. Beyond predictive accuracy, the models also identify the features that most significantly influence resale value, which enhances the transparency and guiding buying and refurbishment strategies. Ultimately, this solution empowers businesses to compete more effectively in the smartphone resale market by combining operational efficiency with strategic insights, ensuring that both profitability and inventory objectives are achieved in a balanced, data-driven manner.